Algebra, Monsoon 2026 — Lecture 0

Sets and Mappings

1 Sets and mappings

You may be familiar with *sets* and *mappings*.

- Every mapping f has a *source* (say A) and a *target* (say B); we write $f : A \to B$.
- Every set A has the *identity* mapping $\mathrm{Id}_A : A \to A$.
- Two mappings $f : A \to B$ and $g : C \to D$ are said to be *composable* if $C = B$, in which case there is a *composite* $g \circ f : A \to D$.
- $f \circ \mathrm{Id}_A = f$ and $\mathrm{Id}_B \circ f = f$ for every $f : A \to B$.
- For any triple of mappings $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Let A, B, C be sets and $f : A \to B$, $g : B \to C$, $h : A \to C$ mappings. We display this information as a triangle

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow^{h} & \downarrow{\scriptstyle g} \\ & & C \end{array}$$

We say that this triangle *commutes* if $h = g \circ f$.

Similarly, a square of mappings

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle g} \\ C & \xrightarrow[k]{} & D \end{array}$$

is said to *commute* if $g \circ f = k \circ h$.

2 Injections, surjections, bijections

Definition 2.1. A mapping $f : A \to B$ is

- *injective* if $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in A$;
- *surjective* if for every $y \in B$, there exists an $x \in A$ such that $f(x) = y$;
- *bijective* if it is both injective and surjective.

Proposition 2.2. *Let $f : A \to B$.*

(i) f is injective if and only if $A = \emptyset$ or there exists $r : B \to A$ such that $r \circ f = \mathrm{Id}_A$.

(ii) f is surjective if and only if there exists $s : B \to A$ such that $f \circ s = \mathrm{Id}_B$.

Discussion. For (ii): if f is surjective, choose, for each $y \in B$, an element $s(y) \in f^{-1}(y)$ (using the axiom of choice); then $f \circ s = \mathrm{Id}_B$. Conversely if such an s exists, then f is surjective since $y = f(s(y))$ for every y.

For (i): if $A \neq \emptyset$, fix $a_0 \in A$. If f is injective, define $r(y) = f^{-1}(y)$ if $y \in \operatorname{im} f$, and $r(y) = a_0$ otherwise; then $r \circ f = \mathrm{Id}_A$. Conversely, if such an r exists then $f(x_1) = f(x_2) \Rightarrow x_1 = r(f(x_1)) = r(f(x_2)) = x_2$, so f is injective; and if $A = \emptyset$, f is injective vacuously. □

Proposition 2.3. *(i) f is injective if and only if $fg_1 = fg_2 \implies g_1 = g_2$ (*left cancellable*);*

*(ii) f is surjective if and only if $g_1 f = g_2 f \implies g_1 = g_2$ (*right cancellable*).*

Definition 2.4. $f : A \to B$ is an *isomorphism* if there exists $g : B \to A$ such that $f \circ g = \mathrm{Id}_B$ and $g \circ f = \mathrm{Id}_A$. If so, g is unique; we write $g = f^{-1}$.

Proposition 2.5. $(f^{-1})^{-1} = f$; $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proposition 2.6. *f is an isomorphism if and only if f is bijective.*

Proposition 2.7. *If $g \circ f$ is injective, then so is f. If $g \circ f$ is surjective, then so is g.*

3 Products

Let E_1, E_2 be sets. The *product* of E_1, E_2 is the set $E_1 \times E_2$ of all pairs (x_1, x_2) where $x_1 \in E_1$ and $x_2 \in E_2$, together with the mappings

$$p_1 : E_1 \times E_2 \to E_1, \qquad p_2 : E_1 \times E_2 \to E_2$$

defined by $p_1(x_1, x_2) = x_1$ and $p_2(x_1, x_2) = x_2$. So the product of E_1, E_2 is the triple $(E_1 \times E_2, p_1, p_2)$.

We have

$$(x_1, x_2) = (y_1, y_2) \iff (x_1 = y_1 \text{ and } x_2 = y_2).$$

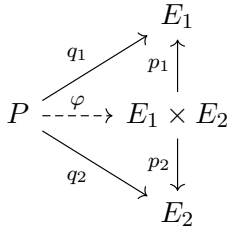
Reformulation. For $z, w \in E_1 \times E_2$,

$$z = w \iff \big(p_1(z) = p_1(w) \text{ and } p_2(z) = p_2(w)\big).$$

Proposition 3.1 (Universal property of the product)**.** *If (P, q_1, q_2) is a triple consisting of a set P and mappings $q_1 : P \to E_1$, $q_2 : P \to E_2$, then there is a unique mapping $\varphi : P \to E_1 \times E_2$ such that*

$$q_1 = p_1 \circ \varphi, \qquad q_2 = p_2 \circ \varphi.$$

Equivalently, there is a unique mapping φ making the two triangles



commute.

Proof. Given (P, q_1, q_2), define φ by $\varphi(x) = (q_1(x), q_2(x))$ for $x \in P$. We clearly have $q_1 = p_1 \circ \varphi$ and $q_2 = p_2 \circ \varphi$.

If $\varphi' : P \to E_1 \times E_2$ is another mapping such that $q_1 = p_1 \circ \varphi'$ and $q_2 = p_2 \circ \varphi'$, then $\varphi' = \varphi$ because for every $x \in P$, the pairs $\varphi'(x), \varphi(x)$ have the same first coordinates (explicitly, $p_1(\varphi'(x)) = q_1(x) = p_1(\varphi(x))$) and the same second coordinates ($p_2(\varphi'(x)) = q_2(x) = p_2(\varphi(x))$). □

Proposition 3.2. *If a triple (P, q_1, q_2) consisting of a set P and mappings $q_1 : P \to E_1$, $q_2 : P \to E_2$ satisfies the same universal property as the triple $(E_1 \times E_2, p_1, p_2)$, then the unique mapping $\varphi : P \to E_1 \times E_2$ such that $q_1 = p_1 \circ \varphi$ and $q_2 = p_2 \circ \varphi$ is a bijection.*

Proof. By the universal property of (P, q_1, q_2), there is a unique mapping $\psi : E_1 \times E_2 \to P$ such that $p_1 = q_1 \circ \psi$ and $p_2 = q_2 \circ \psi$. It is enough to show that $\psi \circ \varphi = \mathrm{Id}_P$ and $\varphi \circ \psi = \mathrm{Id}_{E_1 \times E_2}$.

Indeed, $\psi \circ \varphi : P \to P$ is a mapping satisfying

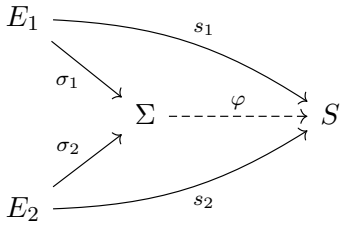
$$q_1 \circ (\psi \circ \varphi) = (q_1 \circ \psi) \circ \varphi = p_1 \circ \varphi = q_1,$$

and similarly $q_2 \circ (\psi \circ \varphi) = q_2$. But we already have the mapping $\mathrm{Id}_P : P \to P$ satisfying $q_1 \circ \mathrm{Id}_P = q_1$ and $q_2 \circ \mathrm{Id}_P = q_2$. By the universal property of (P, q_1, q_2), we get $\psi \circ \varphi = \mathrm{Id}_P$. Similarly, $\varphi \circ \psi = \mathrm{Id}_{E_1 \times E_2}$. Hence φ is a bijection. □

Remark. The pair $(x_1, x_2) \in E_1 \times E_2$ could have been defined using the universal property as follows. Take $q_1 : \{\emptyset\} \to E_1$, $q_2 : \{\emptyset\} \to E_2$ to be the mappings $q_1(\emptyset) = x_1$, $q_2(\emptyset) = x_2$. Then $(x_1, x_2) = \varphi(\emptyset)$.

4 Coproducts

Problem 4.1. Let E_1, E_2 be sets. Does there exist a triple $(\Sigma, \sigma_1, \sigma_2)$ consisting of a set Σ and mappings $\sigma_1 : E_1 \to \Sigma$, $\sigma_2 : E_2 \to \Sigma$ having the universal property that for every triple (S, s_1, s_2) consisting of a set S and mappings $s_1 : E_1 \to S$, $s_2 : E_2 \to S$, there exists a unique mapping $\varphi : \Sigma \to S$ making the two triangles



commute in the sense that $s_1 = \varphi \circ \sigma_1$ and $s_2 = \varphi \circ \sigma_2$?

It can be shown that such a triple $(\Sigma, \sigma_1, \sigma_2)$ does exist; it is unique up to a unique isomorphism. The mappings σ_1, σ_2 are injections. Here is one construction:

$$\Sigma = F_1 \cup F_2, \qquad F_1 = \{1\} \times E_1, \quad F_2 = \{2\} \times E_2,$$

$$\sigma_1(x) = (1, x), \qquad \sigma_2(y) = (2, y) \qquad \text{for every } x \in E_1,\ y \in E_2.$$

We write $\Sigma = E_1 \sqcup E_2$ and call it the *coproduct*, or the *disjoint union*, of the sets E_1, E_2.

5 Equivalence relations

Definition 5.1. Let E be a set. A *relation* on E is a subset $R \subset E \times E$. We often write $x \sim_R y$ to say that $(x, y) \in R$. We say that R is

- *reflexive* if $x \sim_R x$ for all $x \in E$;
- *symmetric* if $x \sim_R y \implies y \sim_R x$ for $x, y \in E$;
- *transitive* if ($x \sim_R y$ and $y \sim_R z$) $\implies x \sim_R z$ for $x, y, z \in E$.

A relation is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Example 5.2. For every mapping $f : E \to F$, the relation $x \sim y \iff f(x) = f(y)$ is an equivalence relation on E.

Example 5.3. Fix an integer $n \in \mathbb{Z}$. The relation $\equiv_n$ on $\mathbb{Z}$ defined by

$$a \equiv_n b \iff a - b = nk \text{ for some } k \in \mathbb{Z}$$

is an equivalence relation. We often write $a \equiv b \pmod{n}$ for $a \equiv_n b$ and say that a is *congruent to b modulo n*.

Definition 5.4. Let E be a set. A *partition* of E is a set P of subsets of E such that

1) $\emptyset \notin P$;

2) $S, T \in P \implies (S = T \text{ or } S \cap T = \emptyset)$;

3) $\bigcup_{S \in P} S = E$.

(The empty set is not in P, any two distinct sets in P are disjoint, and the union of all sets in P is E.)

Example 5.5. If $E \neq \emptyset$, then $P = \{E\}$ is a partition of E. If $E = \emptyset$, then $P = \emptyset$ is a partition of E.

For every $x \in E$, there is a unique $S \in P$ such that $x \in S$. So P gives rise to a surjective mapping $\pi_P : E \to P$ sending each $x \in E$ to the unique $S \in P$ such that $x \in S$.

Definition 5.6. Let E be a set, R an equivalence relation on E, and $x \in E$. The *R-equivalence class* of x is the set $[x]_R$ of all $y \in E$ such that $x \sim_R y$.

Remark. In other words, $[x]_R = p_{2*}\, p_1^*(x)$, where $p_1, p_2 : R \to E$ are the restrictions to R of the two canonical projections $E \times E \to E$. It so happens that, restricted to R, we have $p_{1*}p_2^* = p_{2*}p_1^*$.

The set of R-equivalence classes is denoted E/R.

Proposition 5.7. *Let E be a set and R an equivalence relation on E. The set E/R of R-equivalence classes of elements of E is a partition of E.*

Proof. First, the R-equivalence class $[x]_R$ of an element $x \in E$ cannot be empty because $x \in [x]_R$ (R is reflexive). This fulfills the first condition to form a partition

For an equivalence class to form a partition, either the intersection between equivalence classes are empty, or the sets are equal. Let us assume that the intersection between two equivalence classes are non-empty, i.e. if there is a $z \in [x]_R \cap [y]_R$, then the following holds:

$$z \sim_R x, \qquad z \sim_R y,$$

and so do the following:

$$x \sim_R z \quad \text{(symmetry)}, \qquad y \sim_R z \quad \text{(symmetry)},$$

Now let us consider $x' \in [x]_R$, then

$$\begin{aligned} x' \sim_R x &\implies x' \sim_R z \\ &\implies x' \sim_R y \\ &\implies x' \in [y]_R \\ &\therefore [x]_R \subset [y]_R \end{aligned}$$

By following the same set of steps with $y' \in [y]_R$, we can derive $[y]_R \subset [x]_R$ and therefore $[x]_R = [y]_R$.

Finally, E is clearly the union of all R-equivalence classes. This proves that the set of R-equivalence classes is a partition of E. □

Proposition 5.8. *The equivalence relation on E associated to the canonical surjection $E \to E/R$ is R.*

Proposition 5.9. *Let E be a set. For every partition P of E, there is a unique equivalence relation R on E such that $P = E/R$.*

Proof. Given P, take R to be the equivalence relation on E defined by the surjection $\pi : E \to P$. Then clearly $E/R = P$. We have already seen that if S is an equivalence relation on E such that $E/S = P$, then $R = S$. □

Proposition 5.10. *Let E be a set. The mapping*

$$\varphi : (\textit{equivalence relations on } E) \longrightarrow (\textit{partitions of } E), \qquad R \longmapsto E/R$$

sending an equivalence relation R to the set E/R of R-equivalence classes is a bijection.

Proof. Indeed, the mapping ψ sending a partition of E to the equivalence relation $\psi(P)$ on E defined by

$$x \underset{\psi(P)}{\sim} y \iff \text{there exists an } S \in P \text{ such that } x \in S \text{ and } y \in S$$

is a reciprocal mapping of φ. Indeed,

- for every partition P, the set of $\psi(P)$-equivalence classes is P, so $\varphi \circ \psi = \mathrm{Id}$;
- for every equivalence relation R, the equivalence relation corresponding to the partition $\varphi(R)$ is R, so $\psi \circ \varphi = \mathrm{Id}$. □

Definition 5.11. Let E be a set and R an equivalence relation on E. The set E/R of R-equivalence classes, together with the canonical surjection $\pi : E \to E/R$, $x \mapsto [x]_R$, is called the *quotient* of E by R. Note that

$$x \sim_R y \iff \pi(x) = \pi(y).$$

Example 5.12. Let $\Delta \subset E \times E$ be the diagonal; it is an equivalence relation on E. Show that the canonical surjection $\pi : E \to E/\Delta$ is a bijection.

6 Two lemmas about surjections, and the universal property of the quotient

Lemma 6.1. *Every surjection $\varphi : A \to B$ admits a* section*, i.e., a mapping $s : B \to A$ such that $\varphi \circ s = \mathrm{Id}_B$.*

Proof. For every $y \in B$, the inverse image $\varphi^{-1}(y)$ is not empty, by surjectivity. Choosing an $s(y) \in \varphi^{-1}(y)$ defines a section, as claimed. □

Lemma 6.2. *Consider three mappings*

$$A \xrightarrow{\varphi} B \underset{\psi_2}{\overset{\psi_1}{\rightrightarrows}} C.$$

If φ is surjective and $\psi_1 \circ \varphi = \psi_2 \circ \varphi$, then $\psi_1 = \psi_2$.

Proof. Let $s : B \to A$ be a section of φ. Now,

$$\begin{aligned} \psi_1 \circ \varphi = \psi_2 \circ \varphi &\implies (\psi_1 \circ \varphi) \circ s = (\psi_2 \circ \varphi) \circ s \\ &\implies \psi_1 \circ (\varphi \circ s) = \psi_2 \circ (\varphi \circ s) \\ &\implies \psi_1 \circ \mathrm{Id}_B = \psi_2 \circ \mathrm{Id}_B \\ &\implies \psi_1 = \psi_2. \end{aligned}$$

□

Exercise 6.3. If $\varphi : A \to B$ is an injection, then either $A = \emptyset$ or φ admits a *retraction*, namely a mapping $r : B \to A$ such that $r \circ \varphi = \mathrm{Id}_A$.

Exercise 6.4. Consider a diagram

$$A \underset{\psi_2}{\overset{\psi_1}{\rightrightarrows}} B \xrightarrow{\varphi} C.$$

If φ is injective and $\varphi \circ \psi_1 = \varphi \circ \psi_2$, then $\psi_1 = \psi_2$.

Proposition 6.5 (Universal property of the quotient)**.** *Let E be a set, R an equivalence relation on E, and $\pi : E \to E/R$ the quotient of E by R. For every mapping $f : E \to F$ such that*

$$x \sim_R y \implies f(x) = f(y),$$

there exists a unique mapping $\overline{f} : E/R \to F$ such that $f = \overline{f} \circ \pi$. In other words, there is a unique $\overline{f}$ making the triangle

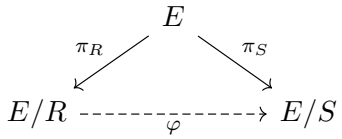
$$\begin{array}{ccc} E & \xrightarrow{\pi} & E/R \\ & \searrow^{f} & \downarrow{\scriptstyle \overline{f}} \\ & & F \end{array}$$

commute.

Proof. First note that the hypothesis $x \sim_R y \implies f(x) = f(y)$ means that $f(x)$ depends only on $\pi(x)$, so we may pose $\overline{f}(\pi(x)) = f(x)$. As π is surjective, this defines a mapping $\overline{f} : E/R \to F$, and $f = \overline{f} \circ \pi$ by construction.

As for the uniqueness of $\overline{f}$: if $g : E/R \to F$ is another mapping such that $f = g \circ \pi$, then $g = \overline{f}$ by Lemma 6.2 (which is applicable because π is surjective). □

Exercise 6.6. Let E be a set, R and S two equivalence relations on E, and $\pi_R : E \to E/R$, $\pi_S : E \to E/S$ the canonical surjections. When does there exist a mapping $\varphi : E/R \to E/S$ making the triangle



commute (in the sense that $\pi_S = \varphi \circ \pi_R$)?

7 The canonical decomposition of a mapping

Let $f : X \to Y$ be a mapping. We have the subset $f_*(X) \subset Y$; denote by $i : f_*(X) \to Y$ the inclusion.

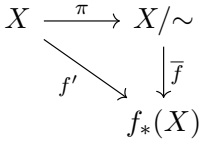
We have the equivalence relation $\sim$ on X,

$$x \sim y \iff f(x) = f(y),$$

and the quotient $\pi : X \to X/\sim$. By the universal property of the quotient, there is a unique mapping

$$\overline{f} : X/\sim \longrightarrow f_*(X)$$

such that the triangle



commutes (where f' is f considered as a (surjective) mapping into $f_*(X)$).

Lemma 7.1. *$\overline{f}$ is a bijection.*

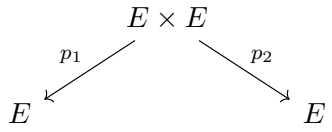
Proof. For every $y \in f_*(X)$, there is an $x \in X$ such that $f(x) = y$; we then have $y = \overline{f}(\pi(x))$, so $\overline{f}$ is surjective.

Let $\overline{x}_1, \overline{x}_2 \in X/\sim$ be such that $\overline{f}(\overline{x}_1) = \overline{f}(\overline{x}_2)$. Then $f(x_1) = f(x_2)$, so $x_1 \sim x_2$, and hence $\overline{x}_1 = \overline{x}_2$, showing that $\overline{f}$ is injective. □

We have seen that $f = i \circ \overline{f} \circ \pi$; this is called the *canonical decomposition* of f. In short, every mapping is a surjection followed by a bijection followed by an injection.

8 Canonical involutions

Let E be a set and let $(E \times E, p_1, p_2)$ be "the" product of E and E,



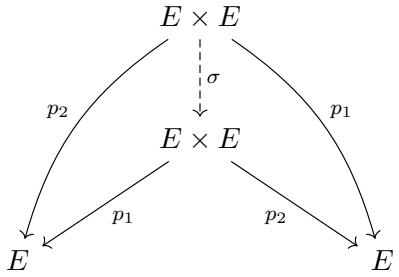
By the universal property of the product, there is a unique morphism of triples

$$\sigma : (E \times E, p_2, p_1) \to (E \times E, p_1, p_2).$$

In other words, there is a unique mapping $\sigma : E \times E \to E \times E$ such that

$$p_2 = p_1 \circ \sigma \qquad \text{and} \qquad p_1 = p_2 \circ \sigma.$$

Equivalently, the two triangles



commute. We have $\sigma(x, y) = (y, x)$ by definition.

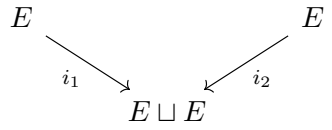
Lemma 8.1. *σ is a bijection.*

Proof. Indeed, σ has an inverse, namely σ. □

Definition 8.2. σ is called the *canonical involution* of $E \times E$.

Remark. σ is an involution of the *set* $E \times E$, not of the *product* $(E \times E, p_1, p_2)$. The only endomorphism of the product is the identity.

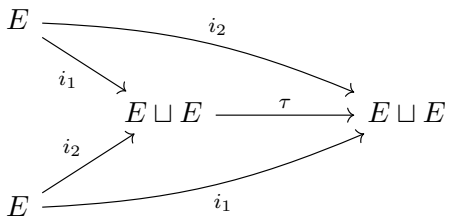
Let E be a set and let $(E \sqcup E, i_1, i_2)$ be "the" coproduct of E and E,



By the universal property of the coproduct, there is a unique morphism of triples

$$\tau : (E \sqcup E, i_1, i_2) \to (E \sqcup E, i_2, i_1).$$

In other words, there is a unique mapping $\tau : E \sqcup E \to E \sqcup E$ such that $i_1 = \tau \circ i_2$ and $i_2 = \tau \circ i_1$. Equivalently, the two triangles



commute. Explicitly, τ is the unique mapping such that

$$\tau(i_1(x)) = i_2(x), \qquad \tau(i_2(x)) = i_1(x) \qquad \text{for every } x \in E.$$

Lemma 8.3. *τ is a bijection.*

Proof. Indeed, τ has an inverse, namely τ. □

Remark. As before, τ is an involution of the *set* $E \sqcup E$, not of the *coproduct* $(E \sqcup E, i_1, i_2)$. The only endomorphism of the coproduct is the identity.

9 Products and coproducts of families

More generally, for any family of sets $(E_\lambda)_{\lambda\in\Lambda}$, we have their *product*, namely a set

$$\Pi = \prod_{\lambda\in\Lambda} E_\lambda,$$

together with mappings $\pi_\lambda : \Pi \to E_\lambda$ $(\lambda \in \Lambda)$ such that, for any pair consisting of a set P and a family of mappings $p_\lambda : P \to E_\lambda$ $(\lambda \in \Lambda)$, there is a unique mapping $\varphi : P \to \Pi$ such that

$$p_\lambda = \pi_\lambda \circ \varphi \qquad \text{for every } \lambda \in \Lambda.$$

Equivalently:

Proposition 9.1. *For every set P, the mapping*

$$\mathrm{Hom}_{\mathbf{Set}}\Big(P, \prod_{\lambda\in\Lambda} E_\lambda\Big) \longrightarrow \prod_{\lambda\in\Lambda} \mathrm{Hom}_{\mathbf{Set}}(P, E_\lambda), \qquad \varphi \longmapsto (\pi_\lambda \circ \varphi)_{\lambda\in\Lambda}$$

is a bijection.

Similarly, the *coproduct* of the family $(E_\lambda)_{\lambda\in\Lambda}$ is a set

$$\Sigma = \coprod_{\lambda\in\Lambda} E_\lambda$$

together with mappings $\iota_\lambda : E_\lambda \to \Sigma$ such that, for every pair consisting of a set S and a family of mappings $\iota'_\lambda : E_\lambda \to S$ $(\lambda \in \Lambda)$, there is a unique mapping $\varphi : \Sigma \to S$ such that $\iota'_\lambda = \varphi \circ \iota_\lambda$ for every $\lambda \in \Lambda$. Equivalently:

Proposition 9.2. *For every set S, the mapping*

$$\mathrm{Hom}_{\mathbf{Set}}\Big(\coprod_{\lambda\in\Lambda} E_\lambda, S\Big) \longrightarrow \prod_{\lambda\in\Lambda} \mathrm{Hom}_{\mathbf{Set}}(E_\lambda, S), \qquad \varphi \longmapsto (\varphi \circ \iota_\lambda)_{\lambda\in\Lambda}$$

is a bijection.

10 The exponential bijection

Proposition 10.1. *There is a bijection*

$$\mathrm{Hom}_{\mathbf{Set}}(E \times F, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}_{\mathbf{Set}}(F, G)\big)$$

given by

$$f \longmapsto \varphi(f), \qquad \varphi(f)(x)(y) = f(x, y),$$

with inverse

$$g \longmapsto \psi(g), \qquad \psi(g)(x, y) = g(x)(y).$$

Proof. We check $\psi \circ \varphi = \mathrm{Id}$ and $\varphi \circ \psi = \mathrm{Id}$:

$$\psi(\varphi(f))(x, y) = \varphi(f)(x)(y) = f(x, y), \qquad \text{so } \psi \circ \varphi = \mathrm{Id};$$

$$\varphi(\psi(g))(x)(y) = \psi(g)(x, y) = g(x)(y), \qquad \text{so } \varphi \circ \psi = \mathrm{Id}.$$

More explicitly:

Injectivity. If $\varphi(f_1) = \varphi(f_2)$, then $\varphi(f_1)(x) = \varphi(f_2)(x)$ for every $x \in E$, so $\varphi(f_1)(x)(y) = \varphi(f_2)(x)(y)$ for every $y \in F$, i.e. $f_1(x, y) = f_2(x, y)$ for every $x \in E, y \in F$; hence $f_1 = f_2$.

Surjectivity. Given $g \in \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathbf{Set}}(F, G))$, define $f(x, y) = g(x)(y)$. Then $\varphi(f) = g$. □

Let Z be a fixed set. For every set X, put $F_Z(X) = X \times Z$; for every set Y, put $h^Z(Y) = \mathrm{Hom}_{\mathbf{Set}}(Z, Y)$. We have come across the bijection

$$\varphi_{X,Y} : \mathrm{Hom}_{\mathbf{Set}}(F_Z(X), Y) \longrightarrow \mathrm{Hom}_{\mathbf{Set}}(X, h^Z(Y)).$$

Notice that:

- for every mapping $a : X' \to X$, we have the mapping

  $$F(a) : F_Z(X') \longrightarrow F_Z(X), \qquad (x', z) \longmapsto (a(x'), z),$$

  i.e. $F(a) = a \times \mathrm{Id}_Z$;
- for every mapping $b : Y \to Y'$, we have the mapping

  $$h^Z(b) : h^Z(Y) \longrightarrow h^Z(Y'), \qquad h^Z(b)(u) = b \circ u.$$

Proposition 10.2. *For every mapping $a : X' \to X$, the square*

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbf{Set}}(F_Z(X), Y) & \xrightarrow{\varphi_{X,Y}} & \mathrm{Hom}_{\mathbf{Set}}(X, h^Z(Y)) \\ {\scriptstyle F(a)\circ(\,\cdot\,)}\downarrow & & \downarrow{\scriptstyle (\,\cdot\,)\circ a} \\ \mathrm{Hom}_{\mathbf{Set}}(F_Z(X'), Y) & \xrightarrow[\varphi_{X',Y}]{} & \mathrm{Hom}_{\mathbf{Set}}(X', h^Z(Y)) \end{array}$$

commutes.

Proposition 10.3. *For every mapping $b : Y \to Y'$, the square*

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbf{Set}}(F_Z(X), Y) & \xrightarrow{\varphi_{X,Y}} & \mathrm{Hom}_{\mathbf{Set}}(X, h^Z(Y)) \\ {\scriptstyle b\circ(\,\cdot\,)}\downarrow & & \downarrow{\scriptstyle h^Z(b)\circ(\,\cdot\,)} \\ \mathrm{Hom}_{\mathbf{Set}}(F_Z(X), Y') & \xrightarrow[\varphi_{X,Y'}]{} & \mathrm{Hom}_{\mathbf{Set}}(X, h^Z(Y')) \end{array}$$

commutes.

11 Correspondences

Let X, Y be sets. A *correspondence* from X to Y is a subset $C \subset X \times Y$.

Example 11.1. $C = \Delta_X$ is a correspondence from X to X.

Define:

$$C^{-1} = \{(y, x) \subset Y \times X \mid (x, y) \in C\}.$$

The above is a correspondence from Y to X.

From the above, we have $(C^{-1})^{-1} = C$.

Let $C \subset X \times Y$ and $D \subset Y \times Z$ be correspondences. Their *composite* is

$$D \circ C = \big\{(x, z) \;\big|\; \text{there exists } y \in Y \text{ such that } (x, y) \in C \text{ and } (y, z) \in D\big\}.$$

We have

$$C \circ \Delta_X = C, \qquad \Delta_Y \circ C = C \qquad \text{for every } C \subset X \times Y,$$

and

$$E \circ (D \circ C) = (E \circ D) \circ C \qquad \text{whenever } C \subset X \times Y,\ D \subset Y \times Z,\ E \subset Z \times W.$$